

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I & II EXAMINATION – WINTER 2024****Subject Code:BE01000041****Date:04-01-2025****Subject Name:Mathematics-I****Time:10:30 AM TO 01:30 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

Q.1 (a) Find $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} - \frac{1}{\sin^2 x} \right)$ **03**

(b) Write second derivative test for local extrema and find the extrema of $f(x) = x^4 - 4x^3 + 10$ **04**

(c) Expand e^x into the power series of x . Find number of terms required to calculate e with an error less than 10^{-4} . **07**

Q.2 (a) Define Beta and Gamma functions and evaluate $\int_0^\infty \frac{x^3(1+x^2)}{(1+x)^{10}} dx$ **03**

(b) Discuss the convergence of $\int_{-\infty}^\infty \frac{1}{1+x^2} dx$ **04**

(c) The circle $x^2 + y^2 = a^2$ is rotated about x -axis to generate a sphere, find its volume and sketch the region. **07**

OR

(c) Find the surface area of a solid generated by the revolution of the circle $r = 2a \cos \theta$ about the initial line. **07**

Q.3 (a) Determine whether $\lim_{(x,y) \rightarrow (0,0)} \frac{x^6 - y^2}{x^3 y}$ exists, find it if exists. **03**

(b) Applying the chain rule find $\frac{dw}{dt}$ if $w = x^2 y - y^2$, where $x = \sin t$ and $y = e^t$ also find $\frac{dw}{dt}$ at $t = 0$ **04**

(c) Find the extreme value of the function $f(x, y) = x^3 + y^3 + 3x^2 - 3y^2$ **07**

OR

Q.3 (a) If $u = \tan^{-1}(y/x)$ prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ **03**

(b) Find the tangent plane and normal line of the surface $f(x, y, z) = x^2 + y^2 + z - 9 = 0$ at point $(1, 2, 4)$ **04**

(c) Applying Lagrange multipliers to determine the dimension of a rectangular box, open at the top, having volume 32 ft^3 , and requiring the least amount of material for its construction. **07**

Q.4 (a) State sandwich theorem for the sequence and discuss the convergence of the sequence $\left\{ \frac{\cos n}{n} \right\}$ **03**

(b) Test the convergence of the series $\sum_{n=0}^\infty \left(\frac{4^n + 3}{5^n} \right)$, also find the sum of the series if it is convergent. **04**

(c) Applying appropriate test find the interval of convergence and radius of convergence of the series $\sum_{n=0}^\infty \frac{(x-5)^n}{n^2}$ **07**

OR

- Q.4** (a) Investigate the convergence of the series $\sum_{n=0}^{\infty} \frac{4^n n! n!}{(2n)!}$ **03**
- (b) Discuss the convergence of the series and find it's sum $\sum_{n=0}^{\infty} \frac{1}{n(n+1)}$ **04**
- (c) Applying appropriate test find the interval of convergence and radius of convergence of the series $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{3^n(n+1)}$ **07**
- Q.5** (a) Evaluate $\iint_R y^2 x \, dA$ over the rectangular region **03**
 $R = \{(x, y) : -3 \leq x \leq 2, 0 \leq y \leq 1\}$
- (b) Use double integration to determine the area of the region R inclosed between the parabola $y = \frac{1}{2}x^2$, and the line $y = 2x$, also sketch the region. **04**
- (c) Applying change of the order of the integration to evaluate $\int_0^2 \int_{\frac{y}{2}}^1 e^{x^2} \, dx \, dy$, also sketch the region. **07**
- OR**
- Q.5** (a) Evaluate $\int_0^1 \int_x^1 \int_0^{y-x} dz dy dx$ **03**
- (b) Evaluate the following integral by changing in to the polar co ordinate. **04**
 $\iint_R e^{x^2+y^2} dA$ where R is the semi circular region bounded by the x-axis and the curve $y = \sqrt{1-x^2}$
- (c) A thin plate covers the triangular region bounded by the x-axis and the line $x = 1$ and $y = 2x$ in the first quadrant. The plate density at the point (x, y) is $\delta(x, y) = 6x + 6y + 6$ find the plate's mass, first moments and the centre of mass about the coordinate axis. **07**

Enrolment No./Seat No _____

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I&II EXAMINATION – SUMMER 2025

Subject Code:BE01000041

Date:05-06-2025

Subject Name:Mathematics-I

Time:10:30 AM TO 01:30 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Investigate the convergence of $\int_0^1 \frac{1}{1-x} dx$.	03
	(b) Define beta and gamma functions. What is the relationship between beta and gamma functions?	04
	(c) The region between the curve $y = \sqrt{x}$, $0 \leq x \leq 4$ and the x -axis is revolved about the x -axis to generate a solid. Find its volume.	07
Q.2	(a) Evaluate $\lim_{x \rightarrow 0} \frac{3x - \sin x}{x}$ using L'Hospital rule.	03
	(b) Find the Taylor series generated by $f(x) = e^x$ at $x = 2$.	04
	(c) Find the Maclaurin series for $f(x) = (1+x)^k$ where k is any real number. Using it, find the Maclaurin series for the function $\frac{1}{1-x}$.	07
	OR	
	(c) Find the local extreme values of the function $f(x) = 3x^4 - 2x^3 - 6x^2 + 6x + 1$.	07
Q.3	(a) Test the series $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$ for convergence or divergence.	03
	(b) Test the convergence of the series $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{n}$.	04
	(c) Define absolutely convergent series and conditionally convergent series. Investigate the convergence of the series $\sum_{n=1}^{\infty} \frac{\sin n}{n^2}$.	07
	OR	
Q.3	(a) Investigate the convergence of the series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$.	03
	(b) Find the sum of the series $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.	04
	(c) For what values of x does the series $\sum_{n=1}^{\infty} \frac{(x-3)^n}{n}$ converge?	07
Q.4	(a) If $f(x, y) = \frac{xy^2}{x^2+y^4}$, does $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ exist?	03
	(b) If $z = x^2y + 3xy^4$, where $x = \sin 2t$ and $y = \cos t$, find $\frac{dz}{dt}$ when $t = 0$.	04
	(c) Find the extreme values of the function $f(x, y) = x^2 + 2y^2$ on the circle $x^2 + y^2 = 1$.	07

OR

- Q.4** (a) Find the equation of the tangent plane at the point $(-2, 1, -3)$ to the ellipsoid $\frac{x^2}{4} + y^2 + \frac{z^2}{9} = 3$. **03**
- (b) Find the directional derivative of the function $f(x, y) = x^2y^3 - 4y$ at the point $(2, -1)$ in the direction of the vector $\mathbf{v} = 2\mathbf{i} + 5\mathbf{j}$. **04**
- (c) Find the local extreme values of the function $f(x, y) = xy - x^2 - y^2 - 2x - 2y + 4$. **07**
- Q.5** (a) Evaluate $\int_0^3 \int_0^2 (4 - y^2) dy dx$ **03**
- (b) Evaluate $\iint_R (3x + 4y^2) dA$ where R is the region in the upper half-plane bounded by the circles $x^2 + y^2 = 1$ and $x^2 + y^2 = 4$. **04**
- (c) Sketch the region of integration, reverse the order of integration and evaluate the integral $\int_0^1 \int_x^1 \sin(y^2) dy dx$. **07**

OR

- Q.5** (a) Evaluate the triple integral $\iiint_B xyz^2 dV$ where B is the rectangular box given by $B = \{(x, y, z) : 0 \leq x \leq 1, -1 \leq y \leq 2, 0 \leq z \leq 3\}$. **03**
- (b) Find the area of the region R bounded by $y = 2x^2$ and $y^2 = 4x$. **04**
- (c) Find the volume of the solid that lies under the paraboloid $z = x^2 + y^2$ and above the region R in the xy -plane bounded by the lines $y = 2x$ and the parabola $y = x^2$. **07**

GUJARAT TECHNOLOGICAL UNIVERSITY**Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025****Subject Code: BE01000041/BE01R00041****Date: 22-12-2025****Subject Name: Mathematics-I****Time: 02:30 PM TO 05:30 PM****Total Marks: 70****Instructions**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Define Improper integral of Second kind and test the convergence of the integral $\int_0^1 \frac{5dx}{(x-1)^2}$	03
(b) Define Gamma function. Prove that $\Gamma(n) = 2 \int_0^\infty e^{-x^2} x^{2n-1} dx$	04
(c) Obtain the Maclaurin's series expansion of $\frac{1}{1-x}$ for $ x < 1$. Using it, expand $\log_e(1-x)$ in powers of x .	07
Q.2 (a) Define Beta function. Show that $\beta(m, n) = \beta(n, m)$	03
(b) Find local extreme value of the function $f(x) = 2x^3 - 6x$	04
(c) Define geometric series and explain its convergence. Test the convergence for the following series. $(i) \sum_{n=1}^{\infty} \frac{1}{4n^2 - 1} \quad (ii) \sum_{n=0}^{\infty} 2^{n/2}$	07
OR	
(c) Define nth term test for the divergence of a series. Test the convergence for the following series. $(i) \sum_{n=1}^{\infty} \frac{1}{n\sqrt{n}} \quad (ii) \sum_{n=1}^{\infty} n \sin\left(\frac{1}{n}\right)$	07
Q.3 (a) Evaluate : $\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} \right)$	03
(b) Find the area generated by revolving the curve $y = x, x \in [0, 1]$	04
(c) Find radius and interval of convergence of the power series $\sum_{n=0}^{\infty} \frac{nx^n}{2^n}$	07

OR

- (a) Evaluate : $\lim_{x \rightarrow 0} \left(\frac{1}{x^2} \right)^{\tan x}$ 03
- (b) Find the volumes of the solids generated by revolving the region under the curve $y = \sqrt{x}$ over the interval $[1, 4]$ about the x-axis. 04

- (c) Define Alternating series. Verify whether the series $\sum_{n=1}^{\infty} \frac{(-1)^n n^2}{n^3 + 1}$ is conditionally convergent or not. 07

- Q.4 (a) Evaluate the limit : $\lim_{(x,y) \rightarrow (0,0)} \frac{\sqrt{x} - \sqrt{y}}{\sqrt{x} + \sqrt{y}}$ 03
- (b) Find Tangent plane and Normal line to the surface $xy^2 + 2yz^2 + 3zx^2 = 6$ at the point $(1,1,1)$. 04
- (c) Find the shortest and the longest distances of the point $(1,1,1)$ from the sphere $x^2 + y^2 + z^2 = 1$ 07

OR

- (a) Discuss the continuity of $f(x, y)$ at the origin. where,

$$f(x, y) = \begin{cases} \frac{2x^2y}{x^4+2y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$
 03
- (b) Find the directional derivative of $f(x, y) = x^2 e^{xy}$ at the point $(1, 0)$ in the direction of $\vec{u} = \hat{i} - \hat{j}$ 04
- (c) Find critical points of the function $f(x, y) = x^3 - y^3 - 3x + 3y + 1$. Also find local maxima, local minima and saddle points, if any. 07

- Q.5 (a) Evaluate : $\iint_R (x + y)^2 dA$; $R = [0, 1] \times [0, 2]$ 03
- (b) Evaluate the integral $\int \int r \sqrt{a^2 - r^2} dr d\theta$ over the region of the upper half of the circle $r = a \cos \theta$ 04
- (c) Evaluate the integral $\int_0^{\infty} \int_x^{\infty} e^{-y^2} dy dx$ by changing the order of the integral. 07

OR

(a) Evaluate $\iint_R dA$ where R is the region bounded by the parabolas $y = x^2$ and $x = y^2$. **03**

(b) Evaluate : $\int_0^1 \int_0^1 \int_{\sqrt{x^2+y^2}}^2 z \, dz dy dx$ **04**

(c) Find the volume of the region bounded by a paraboloid $z = 1 - x^2 - y^2$, a plane $z = 1$ and XY-plane. **07**
